# Telco Customer Churn Analysis

*Final Project Report - DVA Capstone 2*

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## 1. Cover Page

- **Project Title:** Telco Customer Churn Analysis
- **Sector:** Telecommunications
- **Team Members & Roles:**

| Name | Role |
|---|---|
| Aayush Chaturvedi | Member 1 |
| Bhavya Jain | Member 2 |
| Dhruv Sareen | Member 3 |
| Jeevan BR | Member 4 |
| Neel Verma | Member 5 |

- **Institute:** Newton School of Technology
- **GitHub Repository URL:** *Turing_Carambola_Telco-Customer-Churn-Analysis*
- **Tableau Public Dashboard URL:** *Capstone2Dashboard*
- **Submission Date:** April 2026

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## 2. Executive Summary

**Problem:** Customer churn is a critical challenge in telecommunications, where acquiring a new customer costs 5-7× more than retaining an existing one. This project investigates what drives customers to leave a telecom provider and identifies actionable strategies to reduce churn.

**Approach:** Using an IBM Watson sample dataset of 7,043 customers with 21 attributes, we built a full Python-based ETL pipeline to clean, transform, and engineer features from the raw data. We then conducted rigorous exploratory and statistical analysis (Chi-Square tests, Mann-Whitney U tests, ANOVA, Logistic Regression, K-Means clustering) and prepared Tableau-ready exports for interactive dashboards.

**Key Insights:**

1. **26.5% of customers have churned**, putting ~$139,131 in monthly recurring revenue at risk (~$1.67M annualized).
2. **Month-to-month contracts are the #1 churn driver** - 42.7% churn rate vs. 2.8% for two-year contracts.
3. **Electronic check users churn at 45.3%**, far exceeding other payment methods (15-19%).
4. **The first 12 months are the "danger zone"** - new customers churn at 47.4%.

**Key Recommendations:**

1. Launch a contract upgrade incentive program targeting month-to-month customers (potential to save ~$120K/month in revenue at risk).
2. Implement an early engagement onboarding program for the first 12 months of customer tenure.
3. Incentivize automatic payment enrollment to move customers off electronic checks.

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## 3. Sector & Business Context

### Sector Overview

The telecommunications industry is characterized by high competition, low switching costs, and commoditized service offerings. Customer churn rates in the industry typically range from 15-30% annually, making retention a top strategic priority.

### Decision-Maker

This analysis serves **VP-level executives in Customer Retention and Revenue Operations** who need data-driven strategies to reduce churn and protect recurring revenue.

### Why This Problem?

Churn directly erodes revenue and increases customer acquisition costs. Even a 1% reduction in churn rate for a mid-size telecom can translate to millions in preserved revenue annually. Understanding *who* churns and *why* enables targeted, cost-effective interventions.

### Business Value

By identifying the key drivers of churn and segmenting customers by risk level, this analysis enables the business to:- Prioritize retention spend on the highest-risk, highest-value customers - Design targeted campaigns based on specific churn triggers - Forecast revenue at risk with greater precision

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## 4. Problem Statement & Objectives

### Formal Problem Definition

> "What customer demographics, service subscriptions, and account attributes drive customer churn in a telecommunications company, and what targeted retention strategies can reduce the churn rate while maximizing customer lifetime value?"

### Scope

**In Scope:** - Cross-sectional analysis of 7,043 customer records - ETL pipeline in Python (cleaning, feature engineering) - Exploratory Data Analysis with business-oriented insights - Statistical validation (hypothesis testing, regression, clustering) - KPI framework and Tableau dashboard design - Actionable retention recommendations with impact estimates

**Out of Scope:** - Time-series forecasting (dataset is a snapshot, not longitudinal) - Production ML model deployment - Real-time churn scoring - Competitive benchmarking (no competitor data available)

### Success Criteria

1. Identify the top 5 statistically significant drivers of churn
2. Segment customers into distinct risk profiles with differentiated churn rates
3. Deliver 3-5 actionable recommendations with estimated financial impact
4. Produce an interactive Tableau dashboard for stakeholder consumption

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## 5. Data Description

### Source

- **Dataset:** IBM Watson Analytics Sample Dataset
- **Access:** *Telco Customer Churn Dataset*
- **File-name:** WA_Fn-UseC_-Telco-Customer-Churn.csv

### Structure

- **Records:** 7,043 customers
- **Columns:** 21 (raw) → 27 after feature engineering
- **Time Period:** Cross-sectional snapshot (no date column); `tenure` (0-72 months) serves as time proxy

### Key Fields

| Category | Columns | Description |
|---|---|---|
| Identifier | customerID | Unique customer ID |
| Demographics | gender, SeniorCitizen, Partner, Dependents | Customer profile |

| Category | Columns | Description |
|---|---|---|
| Account | tenure, Contract, PaperlessBilling, PaymentMethod | Account characteristics |
| Services | PhoneService, MultipleLines, InternetService, 6 add-ons | Service subscriptions |
| Financials | MonthlyCharges, TotalCharges | Revenue metrics |
| Target | Churn (Yes/No) | 73.5% No, 26.5% Yes |

### Data Limitations & Known Biases

- **No temporal dimension:** Cannot analyze churn trends over time or seasonality
- **Snapshot bias:** Dataset captures one point in time; customers who churned years ago may have different patterns than recent churners
- **No competitive data:** Cannot assess whether churners switched to competitors or simply discontinued service
- **No customer satisfaction data:** Attitudinal drivers (e.g., NPS scores, complaint history) are absent
- **Geographic data absent:** Cannot analyze regional patterns

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## 6. Data Cleaning & ETL Pipeline

All cleaning was performed in Python. Reference: notebooks/02_cleaning.ipynb

### Missing Values

- **11 blank TotalCharges values** detected (stored as empty strings, not NaN)
- All 11 correspond to customers with tenure = 0 (brand-new sign-ups not yet billed)
- **Treatment:** Set to 0.0 - this accurately represents zero lifetime spend

- **Justification:** Dropping these rows would lose valid customer records

### Data Type Conversions

| Column | Before | After | Reason |
|---|---|---|---|
| TotalCharges | String (object) | float64 | Enable numeric analysis |
| SeniorCitizen | Integer (0/1) | String (Yes/No) | Consistency with other categoricals |

### Outlier Detection & Treatment

- **Method:** IQR (Interquartile Range) for tenure, MonthlyCharges, TotalCharges, AvgMonthlySpend
- **Result:** Outliers detected in TotalCharges and AvgMonthlySpend are legitimate high-value customers
- **Decision:** All outliers retained - they represent real customer data, not errors

### Standardization

- Stripped whitespace from all string columns
- Validated all categorical column values for consistency

### Feature Engineering (6 New Columns)

| Feature | Formula | Business Purpose |
|---|---|---|
| AvgMonthlySpend | TotalCharges / max(tenure, 1) | Reveals actual spending pattern |
| NumServices | Count of 6 add-on subscriptions | Measures service bundle depth |
| TenureBucket | Binned: New/Growing/Mature/Loyal | Customer lifecycle segmentation |
| HighValueFlag | MonthlyCharges > 75th percentile | Premium customer identification |

| Feature | Formula | Business Purpose |
|---|---|---|
| HasInternet | InternetService != 'No' | Simplified internet flag |
| ContractRisk | 1 if Month-to-month, else 0 | Binary risk indicator |

### Assumptions

- Blank TotalCharges = $0 spend (new customers)
- Tenure = 0 customers are valid records (not data errors)
- Outliers represent genuine customer behavior

### Output

- **File:** data/processed/telco_churn_cleaned.csv - 7,043 rows × 27 columns, 0 null values

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## 7. KPI & Metric Framework

Reference: notebooks/05_final_load_prep.ipynb

| # | KPI | Formula | Value | Business Relevance |
|---|---|---|---|---|
| 1 | Overall Churn Rate | Churned / Total x 100 | 26.5% | Primary health metric for executive reporting |
| 2 | Monthly Revenue at Risk | Sum of MonthlyCharges (churned) | $139,131 | Direct financial impact of churn |
| 3 | Annualized Revenue Loss | Monthly x 12 | $1,669,570 | Full-year impact for budgeting |
| 4 | ARPU (Overall) | Avg MonthlyCharges | $64.76 | Revenue density benchmark |

| # | KPI | Formula | Value | Business Relevance |
|---|---|---|---|---|
| 5 | ARPU (Churned) | Avg MonthlyCharges (churned) | $74.44 | Churned customers pay MORE |
| 6 | ARPU (Retained) | Avg MonthlyCharges (retained) | $61.27 | Retained customer baseline |
| 7 | Contract Stability Index | % on 1yr+ contracts | 45.0% | Customer commitment level |

**Key KPI Insight:** Churned customers have a higher ARPU ($74.44) than retained customers ($61.27), indicating that price-sensitive, high-paying customers are leaving - a double loss of current revenue and future lifetime value.

### KPI-to-Objective Mapping

- **Churn Rate** → Measures progress toward reducing customer attrition
- **Revenue at Risk** → Quantifies the financial urgency of retention efforts
- **ARPU Comparison** → Reveals whether high-value or low-value customers are at risk
- **Contract Stability** → Tracks success of contract upgrade campaigns

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## 8. Exploratory Data Analysis (EDA)

Reference: notebooks/03_eda.ipynb

### 8.1 Churn Overview

- **26.5%** of customers (1,869 out of 7,043) have churned
- Monthly revenue at risk: **$139,131** (~30.5% of total monthly revenue)
- Churned customers contribute a disproportionately large revenue share relative to their count

### 8.2 Demographics & Churn

| Factor | Churn Rate | Insight |
|---|---|---|
| Gender (Male) | 26.2% | No significant difference - gender is NOT a churn driver |
| Gender (Female) | 26.9% | |
| Senior Citizen (Yes) | 41.7% | Seniors churn at nearly 2x the rate of non-seniors |
| Senior Citizen (No) | 23.6% | |
| Partner (No) | 33.0% | Unattached customers are more flight-prone |
| Dependents (No) | 31.3% | Household commitment correlates strongly with retention |

**Business Insight:** Senior citizens represent a high-risk demographic segment (41.7% churn). Customers without partners or dependents lack the "stickiness" that household commitments provide, making them 1.5-2× more likely to churn.

### 8.3 Contract & Billing Impact

| Factor | Churn Rate |
|---|---|
| Month-to-month contract | 42.7% |
| One-year contract | 11.3% |
| Two-year contract | 2.8% |
| Electronic check payment | 45.3% |
| Mailed check | 19.1% |
| Auto bank transfer | 16.7% |

| Factor | Churn Rate |
|---|---|
| Auto credit card | 15.2% |
| Paperless billing (Yes) | 33.6% |

**Business Insight:** Month-to-month contracts are the single biggest churn predictor - converting even 10% of these customers to annual plans could save ~$12K/month. Electronic check users churn at 45.3% because they haven't committed to automatic billing - a clear signal of low engagement.

## 8.4 Service Adoption & Retention

- **Fiber optic** customers churn at 41.9% vs. DSL at 19.0% and no-internet at 7.4%
- **More add-on services = dramatically lower churn:** 0 add-ons → ~40%+ churn; 5–6 services → <15%
- **OnlineSecurity** and **TechSupport** are the most protective add-ons - customers with these services churn at roughly half the rate

**Business Insight:** Service bundling is a powerful retention lever. Fiber optic customers, despite paying more, churn at the highest rate - suggesting possible service quality issues or price sensitivity that needs investigation.

## 8.5 Tenure Analysis

| Tenure Bucket | Churn Rate | Customer Count |
|---|---|---|
| New (0–12 months) | 47.4% | 2,186 |
| Growing (13–24) | 28.7% | 1,024 |
| Mature (25–48) | 20.4% | 1,594 |
| Loyal (49–72) | 9.5% | 2,239 |

**Business Insight:** The first 12 months are the “danger zone” - nearly half of new customers churn. If a customer survives the first year, retention probability improves dramatically. Onboarding programs and early engagement campaigns are critical.

### 8.6 Revenue Impact

- Churned customers have **higher monthly charges** (avg $74.44 vs. $61.27) but **lower total charges** (they leave before generating significant lifetime value)
- This creates a “double loss” - current revenue plus unrealized future lifetime value

### 8.7 Correlation Analysis

Top correlations with churn (absolute value):

1. **ContractRisk** (month-to-month) - strongest positive correlation

2. **Tenure** - strongest negative correlation (longer tenure = less churn)

3. **InternetService/HasInternet** - internet customers churn more

4. **MonthlyCharges** - moderate positive (higher price = more churn)

5. **Partner/Dependents** - negative (household = retention)

### 8.8 Multi-Factor Risk Profile - “The Perfect Storm”

Customers matching ALL of: Month-to-month + Fiber optic + ≤1 add-on + tenure ≤12 months churn at **2–3× the overall average rate**. This segment should be the #1 priority for retention campaigns.

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## 9. Statistical Analysis

Reference: notebooks/04_statistical_analysis.ipynb

### 9.1 Chi-Square Tests of Independence

Tested all 19 categorical variables against Churn ($\alpha = 0.05$).

**Top associations by Cramer’s V (effect size):**

| Variable | Cramer's V | Effect Size | Significant? |
|---|---|---|---|
| Contract | ~0.39 | Medium | Yes ($p < 0.001$) |
| TenureBucket | ~0.33 | Medium | Yes ($p < 0.001$) |
| OnlineSecurity | ~0.29 | Small-Medium | Yes ($p < 0.001$) |
| TechSupport | ~0.28 | Small-Medium | Yes ($p < 0.001$) |
| InternetService | ~0.28 | Small-Medium | Yes ($p < 0.001$) |
| PaymentMethod | ~0.27 | Small-Medium | Yes ($p < 0.001$) |
| gender | ~0.01 | Negligible | No |

**Interpretation:** Contract type has the strongest statistical association with churn. Gender and PhoneService show negligible association - confirming they are not meaningful churn drivers.

## 9.2 Mann-Whitney U Tests

All numeric variables show statistically significant differences ($p < 0.05$) between churned and retained customers: - **Tenure:** Churned customers have dramatically lower tenure - **MonthlyCharges:** Churned customers pay MORE on average - **TotalCharges:** Churned customers have lower total charges (early departure)

## 9.3 ANOVA - Monthly Charges Across Contract Types

- F-test confirms statistically significant differences in monthly charges across contract types ($p < 0.001$)
- Post-hoc Bonferroni-corrected pairwise comparisons all significant

## 9.4 Logistic Regression

- **Training Accuracy:** ~80%
- **Testing Accuracy:** ~80%

- **ROC-AUC Score:** ~0.84
- **Top churn-increasing features:** Contract type (MTM), internet service, monthly charges
- **Top retention features:** Tenure, number of services, long-term contracts

*Note: This is an analytical model for understanding variable importance, not a production prediction system.*

### 9.5 K-Means Customer Segmentation (k=4)

| Cluster | Count | Avg Tenure | Avg Monthly | Churn Rate | Risk Level |
|---|---|---|---|---|---|
| 0 | 1,899 | 56.6 mo | $86.40 | 10.0% | Medium-Low |
| 1 | 2,094 | 9.2 mo | $48.82 | 42.0% | Medium-High |
| 2 | 1,268 | 40.0 mo | $27.31 | 3.0% | Low |
| 3 | 1,782 | 28.4 mo | $87.08 | 44.0% | High |

**Interpretation:** - **Cluster 3 (High Risk):** 100% month-to-month, high monthly charges, moderate tenure - these are premium customers who haven't committed. Revenue at risk: $70,490/month. - **Cluster 1 (Medium-High):** Very short tenure, 100% month-to-month, low service adoption - new customers who haven't been engaged. - **Cluster 2 (Low Risk):** Long tenure, low charges, committed contracts - the ideal loyal customer profile.

### 9.6 Correlation with Significance Testing

Both Pearson and Spearman correlations confirm all numeric variables are significantly correlated with churn at $p < 0.05$. Tenure (negative) and ContractRisk (positive) are the strongest numeric predictors. ## 10. Tableau Dashboard Design

Reference: tableau/screenshots/ and tableau/dashboard_links.md

## Dashboard Objective

Enable executive and operational decision-makers to monitor churn health, identify at-risk customer segments, and drill down into specific churn drivers - all without requiring technical analysis skills.

## View Structure

**View 1 - Executive Summary:** - KPI cards (churn rate, revenue at risk, ARPU, customer count) - Churn distribution donut chart - Top churn drivers ranked by impact - Churn rate by tenure bucket

**View 2 - Customer Risk Segments:** - K-Means cluster scatter plot (tenure vs. monthly charges, colored by cluster) - Cluster profile comparison table - Filterable customer-level detail table

**View 3 - Service & Contract Deep-Dive:** - Service adoption heatmap with churn rates - Contract type comparison bars - Payment method churn breakdown

**View 4 - Revenue Impact & Retention Targeting:** - Revenue at risk treemap by segment - High-value customer scatter (MonthlyCharges vs. TotalCharges, colored by churn) - Retention action recommendations

## Filters & Interactive Elements

- Contract type filter
- Internet service filter
- Tenure bucket selector
- Cluster/segment filter
- Churn status toggle

## Tableau Public URL - *Capstone2Dashboard*

## Data Sources

| File | Purpose |
|---|---|
| tableau_main.csv | Full cleaned dataset (7,043 rows x 27 columns) |
| tableau_kpi_summary.csv | Pre-aggregated KPI values (9 KPIs) |

| File | Purpose |
| --- | --- |
| tableau_segment_churn.csv | Churn rates by 11 segment dimensions |
| tableau_cluster_profiles.csv | K-Means cluster profiles (4 clusters) |

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## 11. Insights Summary

| # | Insight | Decision Implication |
| --- | --- | --- |
| 1 | 26.5% of customers have churned, putting $139K/month revenue at risk | Churn is a financially material problem requiring immediate strategic attention |
| 2 | Month-to-month contracts churn at 42.7% vs. 2.8% for two-year contracts | Contract upgrade campaigns are the highest-leverage tactic available |
| 3 | Electronic check users churn at 45.3%, far exceeding auto-pay users (15–17%) | Electronic check is a proxy for low engagement - incentivize auto-pay enrollment |
| 4 | Fiber optic customers churn at 41.9% despite paying the highest monthly charges | Investigate fiber service quality; high price + high churn suggests unmet expectations |
| 5 | Customers with 5–6 add-on services churn at <15% vs. 40%+ for 0 add-ons | Service bundling dramatically reduces churn - prioritize bundle upsell campaigns |
| 6 | OnlineSecurity and TechSupport are the most protective add-ons | Include these in retention bundles and consider offering them at a discount |
| 7 | The first 12 months are the “danger zone” - 47.4% churn rate for new customers | Invest heavily in onboarding and early-tenure engagement touchpoints |

| # | Insight | Decision Implication |
|---|---|---|
| 8 | Senior citizens churn at 41.7% vs. 23.6% for non-seniors | Develop senior-specific retention programs with dedicated support |
| 9 | Churned customers pay $13.17 MORE per month than retained customers | Price sensitivity is a factor - justify costs with loyalty pricing/value-add |
| 10 | K-Means clustering reveals 4 segments with churn rates from 3% to 44% | Enable targeted, segment-specific campaigns instead of one-size-fits-all |
| 11 | “Perfect Storm” profile (MTM + Fiber + <=1 addon + new) churns at 2–3x average | This specific segment should receive priority retention intervention |
| 12 | Customers without partners or dependents churn 1.5–2x more than families | Consider household/family bundle offers to increase switching costs |

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## 12. Recommendations

### Recommendation 1: Contract Upgrade Incentive Program

- **Insight:** Month-to-month contracts churn at 42.7% vs. 2.8% for two-year
- **Action:** Offer month-to-month customers a 10–15% discount for switching to annual contracts, with a free month as an enrollment bonus
- **Expected Impact:** Converting 20% of MTM customers (775 customers) to annual contracts could reduce monthly revenue at risk by ~$24K/month ($288K/year)
- **Feasibility:** Low complexity - requires pricing adjustment and marketing campaign; can be implemented within 30 days

### Recommendation 2: Early Engagement Onboarding Program

- **Insight:** 47.4% of customers in their first 12 months churn

- **Action:** Implement a structured 90-day onboarding journey: welcome call at Day 7, service optimization review at Day 30, loyalty offer at Day 90, and quarterly check-ins through Month 12
- **Expected Impact:** Reducing first-year churn by 10 percentage points would retain ~219 additional customers, saving ~$16K/month in revenue
- **Feasibility:** Medium complexity - requires CRM workflow setup and customer success team coordination; 60–90 day implementation

### Recommendation 3: Auto-Pay Migration Campaign

- **Insight:** Electronic check users churn at 45.3% vs. 15–17% for auto-pay users
- **Action:** Offer a $5/month bill credit for customers who switch from electronic check to auto-pay (bank transfer or credit card)
- **Expected Impact:** If 30% of e-check users (709 customers) convert, expected churn reduction of ~200 customers, saving ~$15K/month
- **Feasibility:** Low complexity - billing system change + targeted email/SMS campaign; 30 days to implement

### Recommendation 4: Protective Service Bundling

- **Insight:** OnlineSecurity and TechSupport reduce churn by ~50% among internet users
- **Action:** Create a “Security & Support” bundle at a 25% discount for customers currently without these services, especially targeting fiber optic customers
- **Expected Impact:** Increasing add-on adoption among fiber users could reduce their 41.9% churn rate by 10–15 points, saving ~$30K/month
- **Feasibility:** Medium complexity - requires product packaging and pricing changes; 45–60 days

### Recommendation 5: Senior Citizen Retention Program

- **Insight:** Senior citizens churn at 41.7% (nearly double the average)
- **Action:** Launch a “Senior Care” program: simplified billing, dedicated phone support line, and a 10% loyalty discount for seniors on MTM contracts
- **Expected Impact:** Reducing senior churn by 10 points would retain ~48 additional customers, saving ~$4K/month
- **Feasibility:** Low-medium complexity - requires segment targeting and support channel setup; 45 days

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## 13. Impact Estimation

| Recommendation | Customers Saved | Monthly Saved | Annual Saved |
|---|---|---|---|
| Contract Upgrades | ~775 | ~$24,000 | ~$288,000 |
| Onboarding Program | ~219 | ~$16,000 | ~$192,000 |
| Auto-Pay Migration | ~200 | ~$15,000 | ~$180,000 |
| Service Bundling | ~390 | ~$30,000 | ~$360,000 |
| Senior Retention | ~48 | ~$4,000 | ~$48,000 |
| TOTAL | ~1,632 | ~$89,000 | ~$1,068,000 |

**Combined potential impact:** Implementing all 5 recommendations could reduce the churn rate from 26.5% to approximately 18–20%, recovering up to ~$1.07M in annualized revenue - roughly 64% of the current $1.67M at risk.

**Why act now?** Every month of delay costs an additional ~$139K in lost revenue. The first three recommendations (contract upgrades, auto-pay, onboarding) are low-complexity and can begin within 30 days, capturing early savings while longer-term programs are developed.

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## 14. Limitations

1. **Cross-sectional data only:** The dataset is a single snapshot with no time dimension. We cannot track how churn drivers evolve over time, identify seasonal patterns, or build true time-series churn prediction models.

2. **No causal inference:** All findings are correlational. For example, while electronic check users churn more, we cannot confirm whether the payment method *causes* churn or is merely a proxy for customer engagement level.
3. **Missing attitudinal data:** Customer satisfaction scores (NPS/CSAT), complaint history, and customer service interactions are absent. These subjective factors are often strong churn predictors that could improve model accuracy.
4. **No competitive context:** We do not know whether churned customers switched to a competitor, downgraded services, or discontinued telecom services entirely. This limits our ability to assess competitive positioning.
5. **Class imbalance:** The 73.5%/26.5% split, while not extreme, means the logistic regression model may underpredict churn for borderline cases. For production use, resampling or threshold tuning would be needed.
6. **Feature engineering assumptions:** Setting blank TotalCharges to 0 and treating tenure=0 as valid (rather than data quality issues) are reasonable but unverifiable assumptions.

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## 15. Future Scope

1. **Predictive churn scoring model:** With additional temporal data, build a production-grade ML model (Random Forest, XGBoost, or neural network) to assign real-time churn probability scores to each customer.
2. **Customer lifetime value (CLV) prediction:** Combine churn probability with revenue data to forecast expected CLV, enabling ROI-optimized retention spending.
3. **Real-time dashboard:** Connect the Tableau dashboard to a live data source for ongoing monitoring rather than static snapshot analysis.
4. **A/B testing framework:** Design controlled experiments to test the effectiveness of each retention recommendation (e.g., measuring actual churn reduction from contract upgrade offers).
5. **Additional data sources:** Integrate customer service call logs, NPS survey responses, network quality metrics, and competitive pricing data to build a more complete picture of churn drivers.

6. **Survival analysis:** Apply Cox proportional hazards or Kaplan-Meier survival curves to model time-to-churn more precisely.

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## 16. Conclusion

This project demonstrates that customer churn in telecommunications is a predictable, preventable problem when analyzed with rigorous data methods. Starting from a raw dataset of 7,043 customer records, we built a complete Python-based ETL pipeline, conducted exploratory and statistical analysis validated by hypothesis testing, and identified the key levers of churn: contract type, tenure, internet service, payment method, and service bundling. The analysis reveals that a targeted, five-part retention strategy - focusing on contract upgrades, early engagement, auto-pay migration, service bundling, and senior-specific programs - could recover up to $1.07M in annualized revenue, addressing 64% of the current $1.67M at risk. The most impactful single action is converting month-to-month customers to annual contracts, which alone could save an estimated $288K per year.

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## 17. Appendix

### A. Data Dictionary

See docs/data_dictionary.md in the GitHub repository for full column-by-column definitions of all 27 columns (21 raw + 6 engineered).

### B. Python Notebook Index

| Notebook | Purpose | Key Output |
|---|---|---|
| 01_data_sourcing.ipynb | Dataset validation | Confirmed 7,043 rows, 21 cols |
| 02_cleaning.ipynb | ETL pipeline | telco_churn_cleaned.csv |
| 03_eda.ipynb | Exploratory analysis | 12 vis sets, 10 insights |

| Notebook | Purpose | Key Output |
|---|---|---|
| 04_statistical_analysis.ipynb | Statistical validation | Hypothesis tests & ML models |
| 05_final_load_prep.ipynb | KPI computation & export | 4 Tableau-ready CSV files |

### C. Cleaning Log Summary

| Step | Action | Records Affected |
|---|---|---|
| Blank TotalCharges -> 0.0 | Missing value treatment | 11 rows |
| TotalCharges string -> float64 | Type conversion | 7,043 rows |
| SeniorCitizen 0/1 -> Yes/No | Encoding standardization | 7,043 rows |
| Whitespace stripping | String cleanup | All string columns |
| 6 feature columns added | Feature engineering | 7,043 rows |
| Duplicate check | Validation | 0 duplicates found |

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## 18. Contribution Matrix

| Team Member | **Dataset & Sourcing** | **ETL & Cleaning** | **EDA & Analysis** | **Statistical Analysis** | **Tableau Dashboard** | **Report Writing** | **PPT & Viva** |
|---|---|---|---|---|---|---|---|
| Aayush Chaturvedi | ☐ | ☐ | ☑ | ☑ | ☑ | ☐ | – |

| Team Member | Dataset & Sourcing | ETL & Cleaning | EDA & Analysis | Statistical Analysis | Tableau Dashboard | Report Writing | PPT & Viva |
|---|---|---|---|---|---|---|---|
| Bhavya Jain | ☑ | ☐ | ☐ | ☑ | ☐ | ☑ | – |
| Dhruv Sareen | ☐ | ☑ | ☑ | ☐ | ☐ | ☑ | – |
| Jeevan BR | ☐ | ☑ | ☑ | ☑ | ☐ | ☐ | – |
| Neel Verma | ☑ | ☑ | ☐ | ☐ | ☑ | ☐ | – |

**Note:** All team members contributed equally across all project phases. Each member participated in peer review of all deliverables. Contributions are evidenced in GitHub commit history, PR reviews, and committed files.

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*End of Report*